

Technical Report Writing (TRW)

Chapter No 9
(Techniques)

Defining

Chapter in a Nutshell

- What is Defining?
- Creating Formal Definitions
- Creating Informal Definitions
- Creating Extended Definitions

What is Defining?

- Definitions orient (familiarize) readers and help them place new concepts in context. Definitions also explain new terms and concepts to. Since we denote items, concepts, phenomena, system, process, machine, parts, functions with a word or a term, therefore by defining that word or term which denotes something, we define that actual thing, that item, concept, phenomena, system, process, machine, part and function and so on.

Creating Formal Definitions

- A formal definition is one sentence that contains three parts:
 - (a) The term itself that needs defining
 - (b) The class to which the term (item) belongs, aka classification
 - (c) To differentiate the term from other members of its assigned class

To classify the term

To define a term, you first classify the term by placing it in a class or that large group to which the term belongs. This class can be either broad or narrow. For example, a pen can be classed as a 'thing' or as a 'writing instrument'. A carburetor can be classed as a 'part' or as a 'mixing chamber'. The narrower the class we assign, the more meaning is conveyed, and the less is needed to be said in the 3rd part, differentiation. However, the class must be broad enough to be included in the reader's existing knowledge base and they must be able to relate it to something they know.

To differentiate the term

To differentiate the term, explain those characteristics that belong only to it and **NOT** to the other members of the class. If the differentiation applies to more than one member of the class, the definition is imprecise. For example if the writer says, 'Evaporation is the process in which water disappears from a certain area.' Now the differentiation section defines it too broad a sense, because water can disappear from a surface for many reasons, not just from evaporation. The differentiation must explain the characteristics of evaporation that make it unlike any other process, like, 'It is a process of the change of a substance from a liquid to a vapor.'

Differentiate a term with these four common methods:

- Name its essential properties
- Explain what it does
- If the term is an object, describe what it is made of and what it looks like.
- If the term is a process, explain what happen in that process

Creating Informal Definitions

The writer can create informal definitions to facilitate the reader. These are categorized into two types:

1. **Operational definitions:** Operational definition is an agreed upon criterion between two parties for one particular time and one particular place only. It is not universal and is used by organizations, marketers and law enforcement agencies (LEAs) for time being or for keeping things confidential.
2. **Synonyms:** It is a one word replacement only for a word or term under defining. It is effective only when it is easier than the actual term, otherwise it only confuses.

Examples: Despot---Autocrat--- Dictator

Extended Definitions

These are expanded explanations of the terms to be defined. There are eight kinds of extended definitions:

- i. **Explain the derivation:** The writer explains the origin of the term that from where has the term been derived.
- ii. **Explicate the term:** When the writer explains or defines the difficult words contained in the formal definition of a term, it is explicating the term.
- iii. **Use an example:** It helps understand a concept or a term concretely.
- iv. **Use an analogy:** An analogy points out a similarity between otherwise dissimilar things. Comparing something the readers do not know with something they know helps them understand a concept in a better way.
- v. **Compare and contrast:** The writer shows both the similarities and the dissimilarities between the similar things.
- vi. **Explain cause and effect:** Explaining things in their cause and effect relationship enhances understanding.

vii. **Use a visual aid:** Drawings, diagrams, maps, charts, tables etc. reinforce the understanding of concepts and enhance definitions.

viii. **Analyze the term:** The writer divides a term into its parts and helps readers understand it in bits.

Thank you!